

# Software Architecture Documentation: A Survey

Thank you so much for considering to participate in our study!

We are a group of researchers from the the University of Florence (Italy) and the Vrije Universiteit Amsterdam (The Netherlands).

We are seeking feedback on **how architecture views are used in Open Source Software (OSS)**.

An **architecture view is a visual representation**, usually boxes and arrows, **of a software system's structure and/or related information**.

In our [previous research](#), we analyzed 15,000 architecture views across OSS projects, revealing insights into their creation, representation, and usage.

With this survey, we want to:

1. Validate our previous findings
2. Understand developers' needs regarding architecture views
3. Identify strategies to improve architecture view practices

The survey takes around 5-10 minutes, would provide us **invaluable feedback**, and **could help the OSS community to see more high quality architecture views around in the future**.

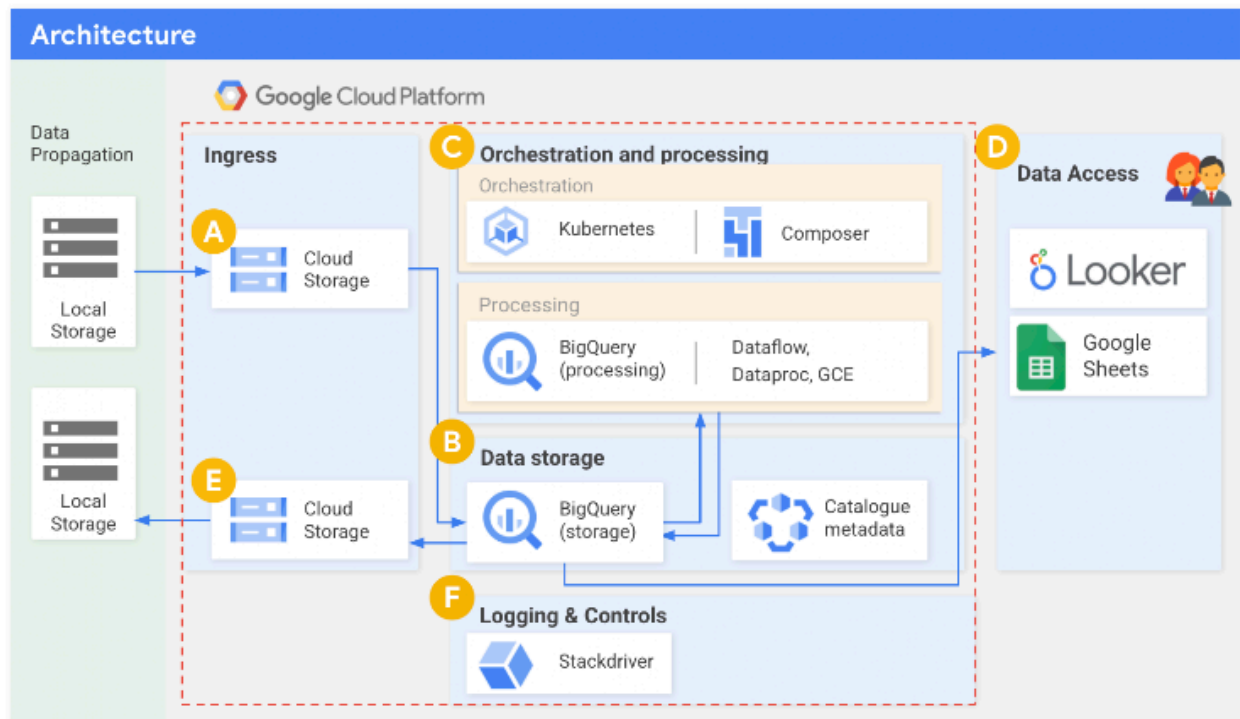
If you wish, we will share with you the final results of the survey. Just specify it at the end of the survey!

**Note:** This survey is exclusively intended for OSS contributors. If you do not contribute to OSS, please do not proceed.

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\* Indicates required question

Example of architecture view taken from [this repository](#).



1. Which of the following sentences **BEST** describes an architecture view? \*

*Mark only one oval.*

- ☐ A detailed code implementation of a software system
- ☐ A graphical representation showing the structure of system components
- ☐ A project management timeline showing development milestones
- ☐ A list of technologies that could be used

**Your experience in software engineering**

2. How many years have you been developing software? \*

*Mark only one oval.*

- ☐ 0-1 years
- ☐ 2-3 years
- ☐ 4-6 years
- ☐ 7-10 years
- ☐ 11-15 years
- ☐ 16+ years

3. Number of OSS projects contributed to with at least one commit \*

*Mark only one oval.*

- ☐ 1-3 projects
- ☐ 4-10 projects
- ☐ 11-20 projects
- ☐ 21+ projects

4. How would you describe your OSS contribution? \*

*Mark only one oval.*

- ☐ Occasional contributor
- ☐ Regular contributor
- ☐ Core/maintainer team member

## 5. What is your current role(s) in your organization? \*

*Check all that apply.*

- ☐ Software Developer
- ☐ Software Architect
- ☐ Product Owner
- ☐ Technical Lead
- ☐ DevOps Engineer
- ☐ Other: \_\_\_\_\_

## 6. What is your primary role in relation to architecture views? \*

*Mark only one oval.*

- ☐ Creating architecture views
- ☐ Using architecture views
- ☐ Both creating and using architecture views
- ☐ Neither

**Current use of architecture views in OSS**

To which extent do you agree with the following statements?

7. Architecture views should use an **informal notation** (for example boxes and arrows, icons) \**Mark only one oval.*

- 1   2   3   4   5
- 
- Stro ☐ ☐ ☐ ☐ ☐ Strongly agree
-

8. **Legends are not necessary** in an architecture view \*

*Mark only one oval.*

|      |                       |                       |                       |                       |                       |                |
|------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------|
|      | 1                     | 2                     | 3                     | 4                     | 5                     |                |
| Stro | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Strongly agree |

9. Architecture views should show the **entire project** and **not just a portion of it** \*

*Mark only one oval.*

|      |                       |                       |                       |                       |                       |                |
|------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------|
|      | 1                     | 2                     | 3                     | 4                     | 5                     |                |
| Stro | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Strongly agree |

10. Architecture views should provide a **high level overview** of a project \*

*Mark only one oval.*

|      |                       |                       |                       |                       |                       |                |
|------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------|
|      | 1                     | 2                     | 3                     | 4                     | 5                     |                |
| Stro | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Strongly agree |

11. **Design information** and **runtime information** are **equally important to be reported** in architecture views \*

*Mark only one oval.*

|      |                       |                       |                       |                       |                       |                |
|------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------|
|      | 1                     | 2                     | 3                     | 4                     | 5                     |                |
| Stro | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Strongly agree |

12. **Technologies** used in a project **are important** in architecture views \**Mark only one oval.*

|      |                       |                       |                       |                       |                       |                |
|------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------|
|      | 1                     | 2                     | 3                     | 4                     | 5                     |                |
| Stro | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Strongly agree |

**Understanding architecture view requirements**13. What is the **most valuable purpose(s)** of an architecture view? \**Check all that apply.*

- ☐ Clarify the system structure
- ☐ Support onboarding new developers
- ☐ Ease understanding complex interactions
- ☐ Assist communication among team members
- ☐ Help in design decision-making
- ☐ Aid system documentation
- ☐ Facilitate system evolution tracking
- ☐ Highlight architecture patterns
- ☐ Visualize runtime behaviour
- ☐ Show component dependencies
- ☐ Other: \_\_\_\_\_

14. What is the **most important concern(s)** an architecture view should cover? \**Check all that apply.*

- ☐ General documentation
- ☐ Deployment
- ☐ Control flow
- ☐ Data flow
- ☐ Connection of components
- ☐ Other: \_\_\_\_\_

15. Which is the **most important quality attribute** an architecture view should cover?

\*

*Check all that apply.*

- ☐ Functionality
- ☐ Performance
- ☐ Compatibility
- ☐ Interaction capability
- ☐ Reliability
- ☐ Security
- ☐ Maintainability
- ☐ Flexibility
- ☐ Safety
- ☐ Not sure

16. For which **activity of the project lifecycle** do you find **architecture views most useful?** (5-point usefulness grid)

\*

*Mark only one oval per row.*

|                                     | Very<br>unimportant   | Unimportant           | Neutral               | Important             | Very<br>important     |
|-------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| <b>Architecture<br/>design</b>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <b>Coding /<br/>implementation</b>  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <b>Maintenance /<br/>operations</b> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

17. Are you using in some way any **standard notation** in your architecture views? If so, which ones?

*Check all that apply.*

- ☐ UML
- ☐ C4
- ☐ 4+1 View Model
- ☐ TOGAF
- ☐ ArchiMate
- ☐ Reference Model of Open Distributed Processing (RM-ODP)
- ☐ IEEE 1471/ISO/IEC 42010
- ☐ Zachman Framework
- ☐ Other: \_\_\_\_\_

18. How **important** are the following **elements** in your architecture views?*Mark only one oval per row.*

|                                                            | Very<br>unimportant   | Unimportant           | Neutral               | Important             | Very<br>important     |
|------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| <b>Architectural components (for example code classes)</b> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <b>Connection/Interaction (for example HTTP calls)</b>     | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <b>Layers (for example the front end)</b>                  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <b>Technologies (for example Kubernetes)</b>               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <b>Nested Components (for example elements of a layer)</b> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <b>Users (for example end users)</b>                       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <b>Legends (for example meaning of the colors used)</b>    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

## 19. If you believe important elements were missing above, please specify them with their importance (as example, code snippets - very important)

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**Supporting architecture view creation**

20. Which **tool** do you use to create architecture views? (if any) \*

*Check all that apply.*

- ☐ Draw.io / Diagram.net
- ☐ AWS Architecture
- ☐ PowerPoint / Keynote / Slides
- ☐ Visio
- ☐ Other: \_\_\_\_\_

21. If you are using a tool, what made you pick it?

*Check all that apply.*

- ☐ Easy to use
- ☐ Rich of features
- ☐ Already used in my development company
- ☐ Frequent updates
- ☐ Integration with other tools
- ☐ Versatile
- ☐ Presence of templates
- ☐ Free to use
- ☐ Supports collaboration
- ☐ Other: \_\_\_\_\_

22. What are the **biggest challenges** you face when creating architecture views? \*

*Check all that apply.*

- ☐ Perceived low value
- ☐ Complex to create
- ☐ Hard to maintain
- ☐ Lack of templates
- ☐ Lack of tools
- ☐ Lack of existing documentation
- ☐ Too time consuming
- ☐ Nobody uses the architecture views afterwards
- ☐ Insufficient architectural design skills
- ☐ Other: \_\_\_\_\_

23. Who is the **intended user** of your architecture views? \*

*Check all that apply.*

- ☐ Myself
- ☐ End users
- ☐ New potential contributors
- ☐ Existing contributors
- ☐ Project maintainers
- ☐ Other: \_\_\_\_\_

24. To which extent is it important to have **versionable architecture views** (version control with change tracking)? \*

*Mark only one oval.*

|     |                       |                       |                       |                       |                       |                |
|-----|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------|
|     | 1                     | 2                     | 3                     | 4                     | 5                     |                |
| Not | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Very important |

25. To which extent is it important to have **collaborative environment integration** for architecture view editing? \*

*Mark only one oval.*

|     |                       |                       |                       |                       |                       |                |
|-----|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------|
|     | 1                     | 2                     | 3                     | 4                     | 5                     |                |
| Not | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Very important |

26. To which extent is it important to have ready to be-used **architecture view templates**? \*

*Mark only one oval.*

|     |                       |                       |                       |                       |                       |                |
|-----|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------|
|     | 1                     | 2                     | 3                     | 4                     | 5                     |                |
| Not | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Very important |

27. To which extent is it important to directly **link architecture view elements to software artifact** (for example code classes)? \*

*Mark only one oval.*

|     |                       |                       |                       |                       |                       |                |
|-----|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------|
|     | 1                     | 2                     | 3                     | 4                     | 5                     |                |
| Not | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Very important |

28. To which extent is it important to be able to **automatically generate architecture views** from the code? \*

*Mark only one oval.*

|     |                       |                       |                       |                       |                       |                |
|-----|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------|
|     | 1                     | 2                     | 3                     | 4                     | 5                     |                |
| Not | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | Very important |

29. Do you use any (semi-)automatic tool for architecture view creation? \*

*Mark only one oval.*

☐ Yes

☐ No

30. If yes, please specify which tool you use

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31. Which of the following aspects **could in your opinion be automated?** \*

*Check all that apply.*

☐ Identification of architecture components

☐ Identification of connections / interaction between components

☐ Architecture view creation

☐ Architecture view update

☐ Architecture view validation

☐ Linking architecture view elements to code artefacts (code classes, technologies, etc)

☐ Other: 

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32. In your opinion, how will the creation of architecture views change in the future?

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33. Any further comments o suggestions?

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34. *If you are interested receiving the results of this survey, fill in your email address.*  
Note: We will use your mail exclusively to share with you the final survey results
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